

Security+ 3.4 · Capacity Planning

Mestre CyberSec · Lesson 3.4 — Capacity Planning (3:53) · [Assistir Vídeo no YouTube](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: PEOPLE + TECHNOLOGY + INFRAESTRUTURA pra demanda. MTR/MTTR medem resposta; MTBF medem confiabilidade (mean time between failure = o quão raro falha).

💡 Key Exam Highlights & Critical Concepts

- Capacidade = gente treinada + hardware + infra estrutural
- MTBF = falha média entre falhas (confiabilidade); MTTR = tempo de resposta

🎯 Exam Tips & Typical CompTIA Scenarios

- 'mean time between failures'=MTBF

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.4 — Capacity Planning)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Capacity Planning

The process of determining and provisioning the people, technology, and infrastructure needed to meet current and future service demand without over- or under-provisioning.

⚠ **Exam Trap:** Na SY0-701, capacidade = PESSOAL treinado + hardware/software + infraestrutura (três pilares). Se a questão citar 'demanda futura' ou 'dimensionar recursos', a resposta é capacity planning, não performance tuning.

• Mean Time Between Failures (MTBF)

The average time a repairable system or component operates before failing; a key measure of reliability — the higher the MTBF, the rarer the failures.

⚠ **Exam Trap:** Gatilho literal: 'mean time BETWEEN failures' = MTBF = CONFIABILIDADE. Não troque por MTTR (resposta/reparo) nem MTTF (item não reparável). MTBF alto = falha rara.

• Mean Time to Repair (MTTR)

The average time required to restore a failed component to full operation, from detection/failure to restoration of service; measures responsiveness.

⚠ **Exam Trap:** MTTR mede VELOCIDADE DE RESPOSTA/REPARO (quanto menor, melhor). Distinção chave da prova: MTTR = 'quanto tempo Leva para ARRUMAR'; MTBF = 'quão RARO é falhar'.

• Mean Time to Failure (MTTF)

The average expected operating time of a NON-repairable device (e.g., fan, disk, power supply) until it must be replaced entirely.

⚠ **Exam Trap:** MTTF = item NÃO reparável (troca-se inteiro); MTBF = item reparável. As questões trocam esses termos para induzir erro — leia se há 'between' ou 'to' no enunciado.

• Availability

The proportion of time a system is operational, commonly calculated as $MTBF / (MTBF + MTTR)$ and expressed as uptime percentage (e.g., 'five nines' = 99.999%).

⚠ **Exam Trap:** A disponibilidade une os dois indicadores: MTBF ↑ e/ou MTTR ↓ aumenta a disponibilidade. Se a pergunta pede 'melhorar uptime', respostas que reduzem MTTR (redundância, resposta rápida) são válidas.

• Personnel Capacity

The 'people' pillar of capacity planning: the number, skill level, and training of staff (help desk tiers, sysadmins, incident response teams) required to meet demand peaks.

⚠ **Exam Trap:** Capacidade na prova não é só hardware — inclui GENTE TREINADA. Cenários com help desk sobrecarregado ou equipe sem treinamento apontam para dimensionamento/expansão de pessoal.

• Infrastructure Capacity

The physical and environmental resources that support services — power, cooling, bandwidth, connectivity, and facility space — which must scale along with technology demand.

⚠ **Exam Trap:** Infraestrutura = energia, refrigeração, banda e espaço físico do datacenter. Se o cenário cita enlace saturado, superaquecimento ou falta de energia, é capacidade de INFRAESTRUTURA, não déficit de hardware do servidor.

• Scalability

The ability of a system to add capacity to handle increased load, either vertically (more resources in a single host / scale up) or horizontally (more hosts / scale out).

⚠ **Exam Trap:** Vertical (scale UP) = aumentar recursos da MESMA máquina; Horizontal (scale OUT) = ADICIONAR mais máquinas. Distinção clássica cobrada nas questões de nuvem e dimensionamento (3.3/3.4).

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A regional bank's network operations team has been logging switch failures for the past 18 months. The CISO wants to justify replacing the existing switches by demonstrating to the board that the current hardware is becoming unreliable. Which metric should the team extract from the logs to BEST support this argument?

- A) Mean Time Between Failures (MTBF)
- B) Mean Time To Repair (MTTR)
- C) Recovery Point Objective (RPO)
- D) Recovery Time Objective (RTO)

QUESTION 2 · EXAM SCENARIO

An e-commerce company's primary database server crashed during a holiday sale, and it took the on-call DBA six hours to fully restore service. Post-incident review revealed that diagnostic tools and documentation were the primary causes of the delay. To improve future incident response, which metric should the team focus on reducing?

- A) Mean Time Between Failures (MTBF)
- B) Mean Time To Repair (MTTR)
- C) Mean Time To Failure (MTTF)
- D) Annualized Rate of Occurrence (ARO)

QUESTION 3 · EXAM SCENARIO

A healthcare SaaS provider is anticipating a 400% increase in concurrent users next quarter due to a new partnership with three large hospital networks. The CTO has been asked to develop a capacity plan. According to CompTIA capacity planning principles, which THREE dimensions must the plan address?

- A) People, technology, and infrastructure
- B) Hardware, software, and networking
- C) Budget, timeline, and scope
- D) Compute, storage, and bandwidth only

QUESTION 4 · EXAM SCENARIO

An SLA guarantees 99.9% availability for a critical web application. The infrastructure team has recorded the following metrics over the last year: MTBF = 8,760 hours and MTTR = 8 hours. Which statement BEST describes whether the system is meeting the SLA?

- A) The system meets the SLA with availability of 99.91%.
- B) The system fails the SLA with availability of only 95%.
- C) The system fails the SLA because MTBF must exceed 10,000 hours.
- D) The system meets the SLA because MTBF exceeds MTTR.

QUESTION 5 · EXAM SCENARIO

A streaming media company experiences traffic spikes during major live sporting events that can reach 10x their normal baseline load. Their current architecture uses fixed-capacity on-premises servers, which become overwhelmed during peak events. Which capacity planning strategy should the architect recommend to handle these predictable but highly variable loads?

- A)** Implement horizontal scalability with cloud-based auto-scaling groups
- B)** Purchase additional fixed-capacity servers equal to peak demand
- C)** Reduce service availability during peak events
- D)** Hardcode rate-limiting rules for all users

QUESTION 6 · EXAM SCENARIO

A 24x7 Security Operations Center (SOC) has been experiencing significant alert fatigue, with analysts missing critical alerts buried among thousands of low-priority notifications. Management has determined that simply buying more monitoring tools will not solve the issue. Which element of capacity planning does this situation primarily reveal as undersized?

- A)** Personnel capacity
- B)** Network bandwidth capacity
- C)** Storage capacity
- D)** Licensing capacity

QUESTION 7 · EXAM SCENARIO

An organization uses consumer-grade SSDs in its logging server that are not designed for continuous write-intensive workloads. After 14 months of operation, multiple drives have failed and cannot be repaired — they must be physically replaced. Which reliability metric is MOST appropriate to track for these non-repairable components?

- A)** Mean Time To Failure (MTTF)
- B)** Mean Time Between Failures (MTBF)
- C)** Mean Time To Repair (MTTR)
- D)** Mean Time To Recovery (MTTR)

QUESTION 8 · EXAM SCENARIO

A financial firm's branch offices report that file transfers to the central data warehouse are timing out during the last hour of each business day. The network team needs to perform capacity planning to address the bottleneck. Which infrastructure component should they FIRST measure to confirm the bottleneck?

- A)** WAN bandwidth utilization between branches and the data center
- B)** CPU utilization on branch file servers
- C)** Disk queue length on the data warehouse storage
- D)** Memory utilization on employee workstations

QUESTION 9 · EXAM SCENARIO

An online learning platform's engineering team must forecast server requirements for the next 24 months based on projected user growth, planned feature rollouts, and seasonal enrollment patterns. Which type of capacity planning activity BEST describes this work?

- A) Proactive capacity planning based on trend analysis and forecasting
- B) Reactive capacity planning after system failures occur
- C) Incident response planning triggered by security events
- D) Disaster recovery planning for site-wide outages

QUESTION 10 · EXAM SCENARIO

A managed service provider (MSP) promises its clients a 99.99% uptime SLA. To properly track compliance with the contractual commitment, which combination of metrics should the provider continuously measure for each client's environment?

- A) MTBF and MTTR, used to calculate availability
- B) RPO and RTO exclusively
- C) AES key length and TLS cipher strength
- D) User count and licensing seats

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — Mean Time Between Failures (MTBF)

MTBF measures the average elapsed time between inherent failures of a system during normal operation — it directly indicates how reliable (or unreliable) a component is. A declining MTBF trend over 18 months would clearly justify replacement. MTTR measures repair time (responsiveness), while RPO and RTO are data-recovery and service-restoration objectives tied to disaster recovery, not component reliability.

Question 2: Option B — Mean Time To Repair (MTTR)

MTTR measures how long it takes to restore a failed system back to operational status — including diagnosis, repair, and recovery. Since the delay was caused by diagnostic tools and documentation during the recovery process, MTTR is the correct metric to focus on reducing. MTBF addresses reliability, MTTF applies to non-repairable components, and ARO estimates annual frequency of a threat event.

Question 3: Option A — People, technology, and infrastructure

CompTIA's capacity planning framework requires evaluating three dimensions: People (sufficient trained staff to operate and support), Technology (tools and applications), and Infrastructure (servers, network, facilities). All three must scale together — adding infrastructure without trained personnel, for example, will still result in outages. The other options miss the 'people' dimension or reduce capacity planning to a single technical layer.

Question 4: Option A — The system meets the SLA with availability of 99.91%.

Availability is calculated as $MTBF / (MTBF + MTTR) = 8,760 / (8,760 + 8) = 8,760 / 8,768 \approx 99.908\%$, which rounds to 99.91% — exceeding the 99.9% SLA requirement. Option B uses incorrect math. Option C confuses a specific threshold that doesn't exist. Option D only compares the two values without computing actual availability.

Question 5: Option A — Implement horizontal scalability with cloud-based auto-scaling groups

Horizontal scalability (scaling out) with cloud-based auto-scaling allows the infrastructure to dynamically add or remove instances based on real-time demand — perfect for predictable but variable spikes like live events. Purchasing fixed peak-capacity servers is wasteful and inefficient for 99% of operating time. Rate-limiting degrades the user experience and doesn't address root capacity issues.

Question 6: Option A — Personnel capacity

Alert fatigue from missed critical alerts directly indicates inadequate personnel capacity (too few trained analysts to triage the alert volume). Capacity planning explicitly requires evaluating the 'people' dimension — no amount of additional technology will compensate for insufficient skilled staff. The other options address technical infrastructure layers, not the human-resource gap that is the root cause here.

Question 7: Option A — Mean Time To Failure (MTTF)

MTTF is used for non-repairable components where failure ends the component's useful life (such as SSDs, light bulbs, or fuses). Since the failed SSDs must be physically replaced rather than repaired, MTTF is the appropriate reliability metric. MTBF applies to repairable systems where the system is restored to service after repair, while MTTR measures the repair duration itself.

Question 8: Option A — WAN bandwidth utilization between branches and the data center

File transfer timeouts occurring during a specific time window strongly suggest network bandwidth saturation on the WAN links connecting branches to the data center. Timeouts are a classic symptom of bandwidth exhaustion, and measuring WAN utilization during the affected window will confirm the hypothesis. CPU, disk, and workstation memory issues would typically manifest as slowness or crashes, not specifically as transfer timeouts tied to a recurring time pattern.

Question 9: Option A — Proactive capacity planning based on trend analysis and forecasting

Forecasting future needs based on growth projections, feature plans, and seasonal patterns is the textbook definition of proactive (anticipatory) capacity planning. It allows the organization to provision resources before demand materializes, avoiding service degradation. Reactive planning only addresses issues after they occur, which is far more costly and disruptive than planning ahead.

Question 10: Option A — MTBF and MTTR, used to calculate availability

Availability = $\text{MTBF} / (\text{MTBF} + \text{MTTR})$. To verify whether the 99.99% SLA is being met, the provider must continuously track both MTBF (how often the system fails) and MTTR (how long recovery takes). RPO/RTO are specific recovery objectives within a disaster recovery plan and don't measure ongoing uptime compliance. Cryptographic key lengths and user counts are unrelated to availability metrics.